

Windows Master Cheatsheet

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1) SYSTEM INFORMATION & DIAGNOSTICS

systeminfo	→ Full system info
msinfo32	→ Detailed hardware + BIOS/UEFI + Secure Boot
winver	→ Windows version
dxdiag	→ DirectX + GPU info
taskmgr	→ Open Task Manager
eventvwr.msc	→ Event Viewer (logs)
perfmon	→ Performance Monitor
resmon	→ Resource Monitor

2) CLEANING & MAINTENANCE

temp	→ Open temp folder
%temp%	→ User temp folder
prefetch	→ Prefetch cache (program startup optimization)
cleanmgr	→ Disk Cleanup
wsreset	→ Reset Microsoft Store cache
recent	→ Recently opened files

3) SYSTEM REPAIR (MUST KNOW)

CMD (Run as Administrator):

sfc /scannow

→ Scan and repair corrupted system files.

DISM /Online /Cleanup-Image /RestoreHealth

→ Repair Windows image when SFC alone is not enough.

chkdsk C: /f /r

→ Scan the disk for errors and bad sectors.

bootrec /fixmbr

bootrec /fixboot

bootrec /scanos

bootrec /rebuildbcd

→ Classic BIOS/MBR boot repair (mainly used from recovery environment).

4) DISK & PARTITION MANAGEMENT

diskmgmt.msc → Disk Management GUI

DISKPART basics:

diskpart

list disk → Show disks

select disk X → Select target disk

list volume → Show volumes

select volume X → Select volume

list partition → Show partitions

shrink desired=200 → Shrink volume by 200 MB (MBR2GPT prep)

clean → Erase partition table (DANGEROUS)

convert gpt → Convert empty disk to GPT

convert mbr → Convert empty disk to MBR

5) BOOT ARCHITECTURE: BIOS vs UEFI

Legacy BIOS:

- Old firmware boot system.
- Uses MBR partition table.
- Supports up to 2TB disks.
- Loads the boot sector from the first 512 bytes of disk.

UEFI (Unified Extensible Firmware Interface):

- Modern replacement for BIOS.
- Uses GPT partition table.
- Uses EFI System Partition (ESP) formatted as FAT32.
- Loads bootloader .EFI files instead of raw boot sectors.
- Stores boot entries in NVRAM (firmware memory).

Important Windows boot files:

- EFI System Partition path:
 \EFI\Microsoft\Boot\bootmgfw.efi → Windows Boot Manager
- Fallback/Generic UEFI path:
 \EFI\Boot\bootx64.efi

bootmgfw.efi:

- Binary file (EFI executable).

- Reads BCD (Boot Configuration Data).
- Decides which OS or boot entry to load.
- Hands control to the Windows loader and then the kernel.

6) REBUILDING UEFI BOOT (NO USB METHOD)

Run CMD as Administrator:

1) Mount the EFI system partition:

```
mountvol S: /s
```

2) Check EFI contents:

```
S:
```

```
dir
```

```
cd EFI\Microsoft\Boot
```

```
dir
```

3) Rebuild Windows boot files on EFI:

```
bcdboot C:\Windows /s S: /f UEFI
```

4) List firmware boot entries:

```
bcdedit /enum firmware
```

5) Remove broken duplicate entries (carefully):

```
bcdedit /delete {GUID}
```

This is what you used when fixing UEFI + Valorant issues.

7) NETWORKING COMMANDS

ipconfig /all → Full IP configuration

ping 8.8.8.8 → Test raw connectivity

ping google.com → Test DNS + connectivity

tracert google.com → Trace route to host

netstat -ano → Show open ports + PID

nslookup domain.com → DNS lookup

arp -a → ARP cache

netsh winsock reset → Reset network stack

8) POWERSHELL ESSENTIALS

Start PowerShell:

```
powershell
```

List processes:

Get-Process

Stop process:

Stop-Process -Id -Force

List services:

Get-Service

Restart service:

Restart-Service -Name "w32time"

Get IP info:

Get-NetIPAddress

Test connection:

Test-Connection google.com

9) WINDOWS KEYBOARD SHORTCUTS

Win + R	→ Open Run
Win + X	→ Power menu (Device Manager, Disk Management, etc.)
Win + E	→ File Explorer
Win + Shift + S	→ Region screenshot
Win + D	→ Show desktop
Win + L	→ Lock screen
Win + Ctrl + D	→ New virtual desktop
Alt + Tab	→ Switch apps
Ctrl + Shift + Esc	→ Task Manager

10) SECURITY & HARDENING (OVERVIEW)

- Keep Windows updated.
- Use Windows Security (Defender) + SmartScreen.
- Do not disable Secure Boot unless needed for testing.
- Be careful with unknown .exe files.
- Use "Run as Administrator" only when necessary.
- Create restore points before deep system changes.

11) NOTES ABOUT BOOT OVERRIDE (WHAT YOU EXPERIENCED)

- Boot Override in BIOS lets you boot once from a specific EFI file.
- It does NOT permanently change NVRAM boot entries.

- That is why your UEFI boot only worked when selecting:
"Boot Override → Windows Boot Manager (KINGSTON ...)" manually.
- After rebuilding NVRAM entries with bcdboot, Windows Boot Manager appears normally.